

Week 1: Tilings — facilitator guide

The session at a glance

One action runs through the whole hour: cover a region exactly with copies of a small piece, with no gaps, no overlaps and nothing sticking out. Three questions come back at every table. Can it be done? Which ones can? When it cannot, why not, for sure? Every table meets an impossibility that is proved by counting two kinds of cells (the two colors of a checkerboard; triangles pointing up and down; nine dotted squares), and the oldest table also meets a construction that works forever (covering a 4-by-4, 8-by-8, 16-by-16, . . . board with one square removed by L-shaped pieces).

Suggested seating. K–1 (two kindergartners and a first grader): the parent. The K–1 sequence is scripted step by step below and needs reading aloud and scribing more than mathematics invented on the spot. Grades 2–3 (the four third graders): one mathematician. Grades 4–5 (two fourth graders and a fifth grader): the other mathematician; this table's arguments, an induction and a counting argument, need the most steering. Swap if you prefer; every table's answers and arguments are complete in this guide.

0:00–0:05	free handling of blocks and paper pieces at the tables; grades 2–5 may cut extra pieces
0:05–0:08	whole-group launch
0:08–0:50	table work (42 minutes)
0:50–0:57	share: one impossibility and its reason from each table
0:57–1:00	pieces back into the trays; packets go home

Every child works with their own copy of the pages and their own pieces. The pages are for doing, not reading: at K–1 the adult reads each problem aloud; at the other tables children read, and the adult rereads aloud when a child is stuck. All boards are printed at the true 1-inch scale of the pattern blocks and paper pieces, except the 8-by-8 board of grades 4–5 (0.8-inch cells, meant for drawing).

Materials and preparation

Pattern blocks, sorted into trays. Launch tray: 13 yellow hexagons, 13 red trapezoids, 39 blue rhombi (one hexagon, one trapezoid and three rhombi per person, adults included); after the launch the rhombi are shared out to the tables. K–1 tray: 40 green triangles, 24 blue rhombi, 8 red trapezoids, 3 yellow hexagons. Grades 2–3 tray: 12 blue rhombi (Problem 10; children take turns). Grades 4–5 tray: 24 blue rhombi (Problem 8 needs 12 per covering; two children build at a time while the third records). If blue rhombi run short, the K–1 table has priority. Keep the purple chevrons and the nonstandard pieces off the tables: every count in this packet assumes triangles, rhombi, trapezoids and hexagons only.

Paper pieces, cut from 1-inch grid paper (about 20 minutes with a paper cutter; children can cut spares during free handling):

- Dominoes, 1 by 2 inches: 160. K–1: 10 per child; grades 2–3 and 4–5: 18 per child; the rest are spares.
- L-pieces (an L of three squares): 36, for grades 4–5 (12 per child; the 8-by-8 board needs 21, so the table pools its pieces). Cut 2-by-2 squares, then snip one cell off each.

- Straight pieces, 1 by 3 inches: 24, for grades 4–5, Problem 7 only.
- Black squares, 1 by 1 inch, from black or dark paper: 20 (two per child). Any flat counter that fits inside a 1-inch square also works.

Other: a pencil per child; crayons in four or more colors at K–1; three colored pencils per child at grades 4–5 (Problem 8).

Printing, single-sided so that boards lie flat: K–1 pages (10 pages), 4 copies; grades 2–3 pages (8 pages), 5 copies; grades 4–5 pages (8 pages), 4 copies; one copy per child and one for the adult at the table. This guide, 3 copies. If the grades 4–5 table wants to build the 8-by-8 board physically rather than draw it, draw an 8-by-8 outline on 1-inch grid paper before the session, or tape together four 4-by-4 boards cut from spare copies of their page 1.

Whole-group launch (two to three minutes)

Everyone, adults too, holds one yellow hexagon, one red trapezoid and three blue rhombi.

1. “Cover the hexagon with your three rhombi. Nothing sticks out, no gaps, no overlaps.” (Thirty seconds.) “Hold it up. Look around: did we all do it the same way?” Two different-looking coverings will be in the room; they are the same covering turned by one notch, and if a child says so, agree. “Two ways. Is there a third? Some of you will find out.”
2. “Now cover the red trapezoid with blue rhombi only.” Let them try for twenty seconds. “Nobody can. Who can say why nobody will ever manage it?” Expected: the trapezoid is three triangles and a rhombus is two. Restate it for the room: “A rhombus always covers two triangles. Two, four, six, never three. So there is no point in trying any longer. That is a reason, and the reason is a count.”
3. “At your table you get boards and pieces. Some coverings can be done: find them, and find how many ways there are. Some cannot: then find a reason that would convince this whole room, like the one we just heard. At ten to the hour each table shows one.” Send the tables off; the adults hand out packets and pieces.

Table K–1

Order and timing (42 minutes). Problem 1, 3 min. Problems 2–3, 7 min. Problems 4–5, 12 min. Problem 6, 5 min. Problem 7, 6 min. Problem 8, 4 min. Problem 9, 5 min. Problem 10 is for a child who finishes early, or for the last minutes. If the table is behind at the 30-minute mark, skip Problem 3 and do only the first board of Problem 9. The heart of this table is Problems 4–6, the up-and-down count, and that is what the table shows at the share.

How to run it. Read one problem aloud, let all three children do it with their own pieces, then write the numbers they tell you in the blanks of their own pages. Ask “how do you know?” only after the recording step. Children may use the printed triangle lines to place pieces; that is intended.

Answers and arguments.

Problem 1. Six green triangles, two red trapezoids, three blue rhombi. (A trapezoid is three triangles and a rhombus is two, so 6, 2 and 3 all say that the hexagon is six triangles.)

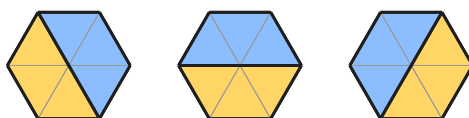
Problem 2. Exactly two ways on the printed picture.



The two rhombus coverings of the hexagon.

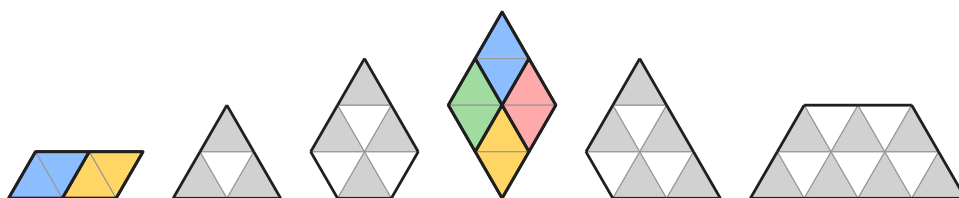
Why only two: every triangle of the hexagon has a corner at the center, so each rhombus has the center as a corner, and the three rhombi meet there. Take the top triangle; inside the hexagon it shares an edge with only two triangles, its left and right neighbors, so its rhombus goes left or goes right. After that choice the other four triangles pair up in only one way. The two ways are the same covering turned by one sixth of a turn; a child who says “it is the same, just turned” is right, and on the fixed printed picture we still count two colorings.

Problem 3. Three ways with trapezoids: the two trapezoids meet along a straight cut through the center joining two opposite corners, and the hexagon has three pairs of opposite corners. One way with triangles. So the counts are: triangles 1, rhombi 2, trapezoids 3.



The three trapezoid coverings.

Problems 4 and 5. A: yes, 2 up and 2 down. B: no, 3 up and 1 down. C: no, 4 up and 3 down. D: yes, 4 and 4. E: no, 5 and 3. F: no, 7 and 5. Children only record yes or no and the two counts here; the reason comes in Problem 6. Shape C has seven triangles, so the launch reason already settles it (rhombi cover 2, 4, 6, ...). Shapes B, E and F have an even number of triangles, so the launch reason is not enough, and that is the point of this page.



Shapes A to F: the coverings of A and D; the up-pointing triangles of the others shaded.

Problem 6. Every placement gives one gray and one white. Reason: a rhombus is two triangles glued along an edge, and two triangles that share an edge always point opposite ways (all three neighbors of an up-pointing triangle point down). The argument for F, in words a five-year-old can repeat: “Every rhombus eats one gray and one white. F has 7 gray and 5 white. After five rhombi all the whites are eaten and two grays are left, and no rhombus can eat two grays.” The same sentence works for B (3 gray and 1 white: after one rhombus two grays are left) and for E (5 and 3). This is the checkerboard argument the older tables use with dominoes, in a different costume: up and down for triangles is black and white for squares. A caution for you only: equal counts do not promise a covering (two triangles that do not touch have equal counts), but every shape on this page with equal counts can be covered, so do not raise this unless a child does.

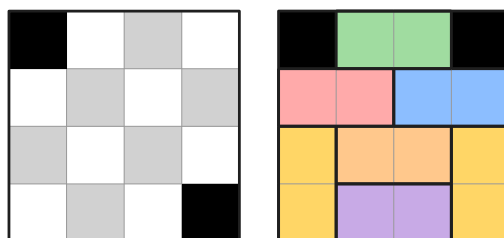
Problem 7. Checks on the four corners and the center; X on the four middle-of-side squares.



Coverings with the black square in a corner and in the center; a middle-of-side square on the checkerboard.

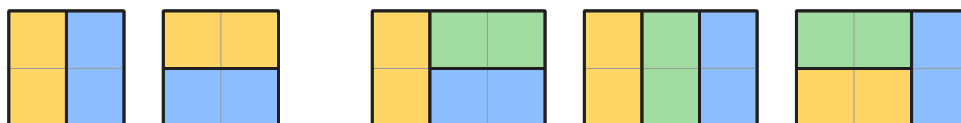
Problem 8. Five dark, four light. Every domino covers one dark and one light square, because two squares that share a side always have different colors. If the black square is on a light square, five dark and three light squares are left; four dominoes would need four light squares; impossible. If the black square is on a dark square, four and four are left, and the coverings above show that it can be done. The check squares of the map are exactly the dark squares.

Problem 9. First board: both black squares are dark (the four corners of a 4-by-4 board have the same color), so 6 dark and 8 light squares remain; seven dominoes would cover 7 dark; impossible. Second board: one dark and one light square are black, 7 and 7 remain, and the board can be covered.



Problem 9: the counts for the first board; a covering of the second.

Problem 10. 2-by-2: two ways. 2-by-3: three ways. Why there is no fourth: the top-left square is covered by a standing domino, and then the remaining 2-by-2 board has two ways, or by a lying domino, and then the square under it must also take a lying domino and the last column is one standing domino, one way. Two plus one.



All coverings of the 2-by-2 and the 2-by-3 board.

Hints, in order. Problem 2: “Where do all three rhombi touch?” then “Turn your hexagon one notch. Does your picture look the same?” Problem 4: “Start at the pointy corner. Which triangle can the corner go with?” (for shape B this forces the first rhombus and leaves two triangles that do not touch). Problem 5: “Put your finger on the point of this triangle. Is the point at the top or at the bottom?” Problem 6: “Try to make your rhombus cover two grays” (they cannot); then “So what does a rhombus always cover?”; then “How many grays in F? How many whites? Pretend each rhombus eats one of each.” Problem 7: “Corner first. Then the middle of a side. Then the center.” Problem 8: “Point to a check on your map. Find that square on the colored board. What color is it?” Problem 9: “What color is each black square?” then “How many dark are left? How many light? Can seven dominoes do that?”

If the table stalls. A child who keeps trying an impossible shape may try for one more minute; then say “three tries and an X, and in a minute we find out why”, and move the group on; the reason in Problem 6 is the reward. If a child cannot tell up from down, put a green block on the printed triangle and ask whether

its point touches the top edge or the bottom edge of its row. If counting the twelve triangles of F is too much, count the gray ones row by row (4, then 3) and the white ones (3, then 2).

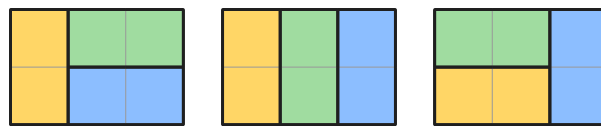
If the table finishes early. Problem 10; then all the ways for a 2-by-4 board on grid paper (five ways); then the 3-by-3 board with the black square in the center: how many different coverings? (Two: the eight remaining squares form a ring, and the four dominoes go around it in one of two ways.) Then a 4-by-4 board with one black square anywhere: can dominoes cover the rest? (Never: fifteen squares is an odd number.)

Table grades 2–3

Order and timing (42 minutes). Problem 1, 4 min. Problem 2, 6 min. Problem 3, 6 min. Problems 4–5, 9 min. Problem 6, 4 min. Problem 7, 6 min. Problem 8, 7 min. Problems 9 and 10 are for children who finish early. If the table is behind at the 30-minute mark, skip Problems 6 and 7: the table should reach the 6-by-6 board, which is what it shows at the share. A table that would rather hold blocks than shade may do Problem 10 before Problem 9.

Answers and arguments.

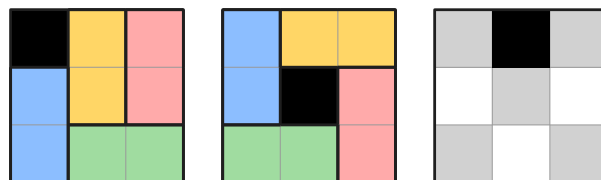
Problem 1. Three ways. Completeness: the top-left square is covered by a standing domino, after which the remaining 2-by-2 board can be covered in two ways, or by a lying domino, after which the square below it must also be covered by a lying domino and the last column by one standing domino, one way. Two plus one is three.



The three coverings of the 2-by-3 board.

Problem 2. Checks on the four corners and the center; X on the four middle-of-side squares.

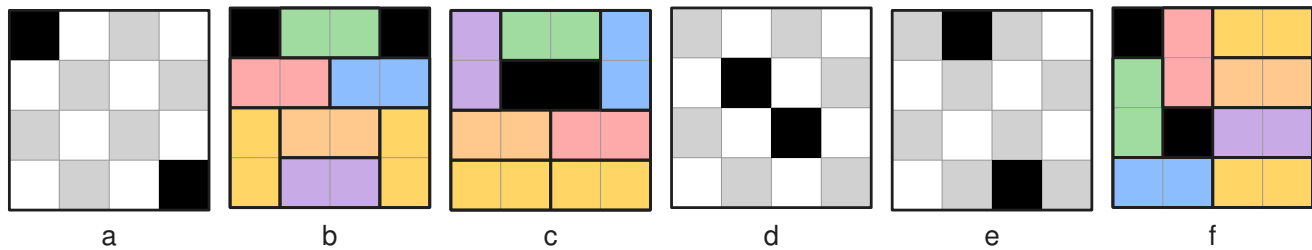
Problem 3. Five shaded (the corners and the center), four unshaded. The checks of Problem 2 are exactly the shaded squares. Every domino covers one shaded and one unshaded square, because two squares that share a side have different shades. The argument the children should produce: “If the black square is on an unshaded square, 5 shaded and 3 unshaded squares are left. Four dominoes cover 4 shaded and 4 unshaded. There are not enough unshaded squares.” With the black square on a shaded square, 4 and 4 are left, and the coverings exist.



Coverings for a corner and for the center (the other corners by turning); the shaded board with a middle-of-side black square.

Problems 4 and 5. a: NO. b: yes. c: yes. d: NO. e: NO. f: yes. Counts of shaded and unshaded squares that are not black, with the top-left square shaded: a 6 and 8, b 7 and 7, c 7 and 7, d 6 and 8, e 8 and 6, f

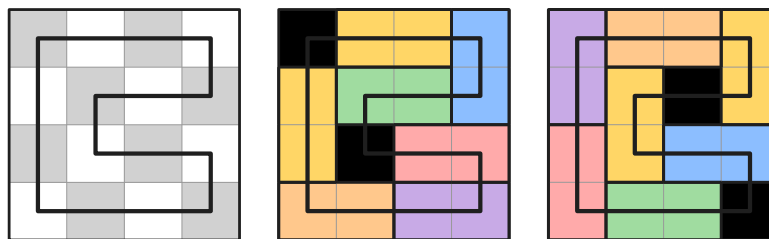
7 and 7. The rule: the two black squares must have different shades, because seven dominoes cover seven shaded and seven unshaded squares. On boards a, d and e the two black squares have the same shade.



Boards a to f: coverings where they exist; the checkerboard where they do not.

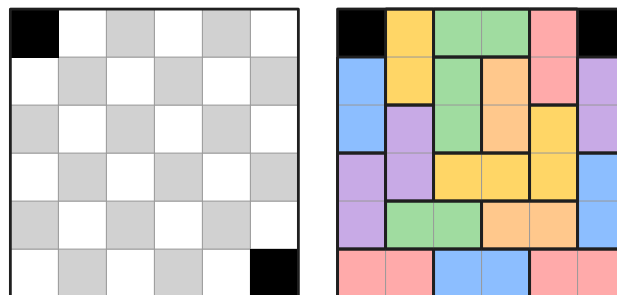
Problem 6. The children will not find a pair of different shades that fails: on the 4-by-4 board, any two squares of different shades can be made black and the rest covered. Problem 7 proves it.

Problem 7. Along a loop through all sixteen squares the shades alternate, shaded, unshaded, shaded, . . . , all the way around, because every step moves to a square of the other shade. Make two squares of different shades black. The loop falls into two stretches. A stretch begins right after one black square and ends right before the other; since shades alternate and the two black squares have different shades, each stretch has an even number of squares (a stretch may be empty, which is fine). Dominoes laid along a stretch, two squares at a time, cover it exactly. Any loop works, so a child's own loop is as good as the one below. On the 8-by-8 board this argument is known as Gomory's theorem.



A loop through all squares, and dominoes laid along it for two pairs of black squares.

Problem 8. Opposite corners: impossible. With the top-left square shaded, both corners are shaded; the full board has 18 shaded and 18 unshaded squares; removing the two corners leaves 16 shaded and 18 unshaded; the 17 dominoes would need 17 of each. Corners of the same side: one shaded and one unshaded square are black, 17 and 17 remain, and the board can be covered: two lying dominoes finish the top row, and the remaining 5-by-6 rectangle is filled row by row with lying dominoes.



The 6-by-6 board: the two cases.

Problem 9. 2-by-4: five ways. 2-by-5: eight ways. So 2, 3, 5, 8, and the expected number for 2-by-6 is 13, which is right. The reason is the one from Problem 1: the leftmost column is covered either by one

standing domino, leaving a board one column shorter, or by two lying dominoes one above the other, leaving a board two columns shorter. So each count is the sum of the two before it: $5 = 3 + 2$, $8 = 5 + 3$, $13 = 8 + 5$. These are the Fibonacci numbers.



The five coverings of the 2-by-4 board.

Problem 10. D: yes, 4 shaded and 4 unshaded. E: no, 5 and 3. F: no, 7 and 5. A rhombus is two triangles sharing an edge, and such triangles always point opposite ways; so each rhombus covers one shaded and one unshaded triangle, exactly as a domino covers one square of each shade. For F, five rhombi would use up the five unshaded triangles and leave two shaded ones, which no rhombus can cover. The coverings of D and the shadings of E and F are in the K–1 figure for shapes A to F above.

Hints, in order. Problem 1: “Look at the top-left square: is its domino standing or lying?” Problem 2: “Corner, then middle of a side, then center; then decide what the other squares are like.” Problem 3: “Can a domino ever cover two shaded squares?” then “After the black square, how many shaded squares are left and how many unshaded? What does each domino take?” Problems 4–5: “Count the shaded and unshaded squares that are left. Could 7 dominoes cover 6 shaded and 8 unshaded?” Problem 6, when a child announces a pair that cannot be covered: “Show me the shades of your two squares” (they will be the same shade, or the covering exists; find it along a loop). Problem 7: if no loop appears within two minutes, draw the loop of the figure on the child’s board; the argument is the point, not the loop. Then: “Walk along the loop from one black square to the other. Count the squares you pass. Odd or even?” Problem 8: “You do not have to try forever. Shade the small board and count.” Problem 9: “Cover the left column first. In how many ways can the left column be covered?”

If the table stalls at Problem 3 and the counting argument does not land: put the black square on a middle-of-side square, cover the three unshaded squares with three dominoes, and ask what the fourth domino is supposed to cover (two shaded squares that do not touch). That is the whole argument in one picture.

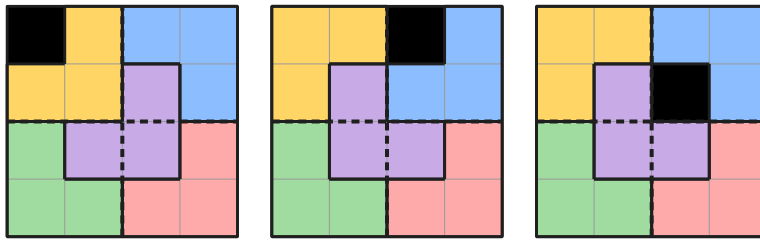
If the table finishes early. Problems 9 and 10. Then: on a 5-by-5 board, which single square can be black so that dominoes cover the rest? (The board has 13 squares of the corners’ shade and 12 of the other; the black square must be on the majority shade, and each of those 13 squares works; for a corner, two lying dominoes finish the top row and the remaining 4-by-5 rectangle is filled row by row.) Then: the 3-by-3 board with the center black has exactly two coverings.

Table grades 4–5

Order and timing (43 minutes). Problem 1, 6 min. Problem 2, 8 min. Problem 3, 8 min. Problem 4, 9 min. Problem 5, 6 min. Problem 6, 6 min. Problems 7 and 8 are for a table that finishes early. The heart is Problems 2–5: the method that always works, and the 5-by-5 board where no method can. If the table is behind at the 30-minute mark, skip Problem 6; the other tables present the checkerboard at the share.

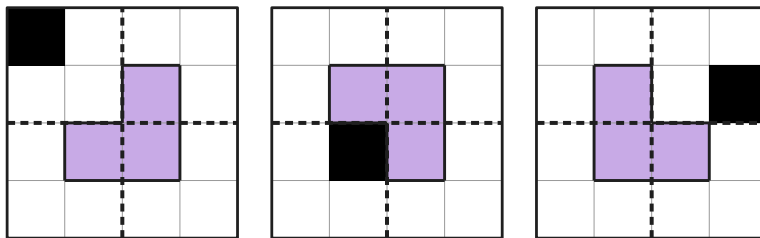
Answers and arguments.

Problem 1. Board A: one L-piece. Board B: five pieces (fifteen squares). For each of the three marked positions there is exactly one covering, so all the children’s drawings will agree with the figure.



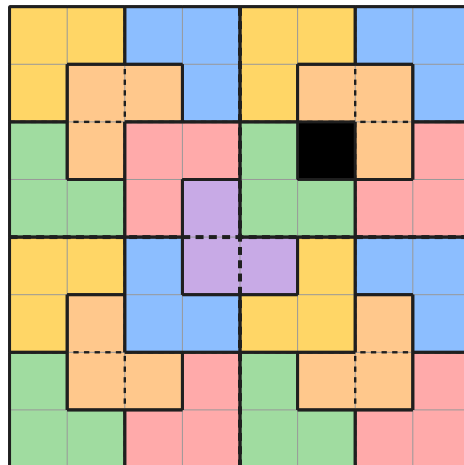
Problem 1: the coverings for the three positions; the dashed lines show the quarters used in Problem 2.

Problem 2. Every position works. The method: the dashed lines cut the board into four 2-by-2 quarters, and the black square lies in one of them. Place one piece in the middle of the board so that it takes one square from each of the three other quarters. Now every quarter is a 2-by-2 board with exactly one square taken, and one piece finishes each quarter; five pieces in all. The method does not care where the black square is, which is what “works for every position” means.



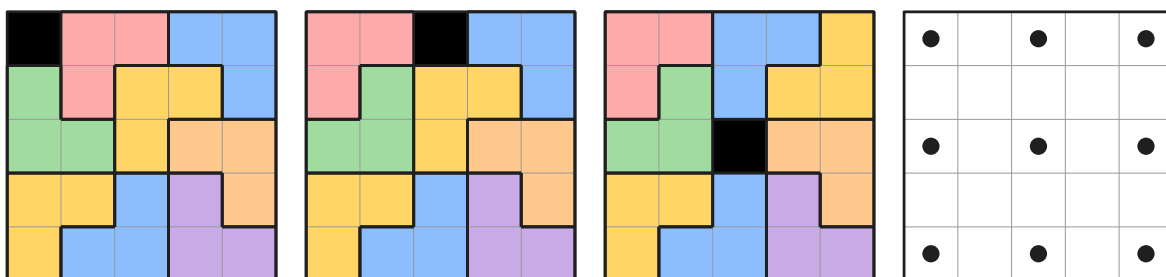
The first piece of the method for three positions of the black square.

Problem 3. Twenty-one pieces (63 squares). The quarters are now 4-by-4 boards. The central piece takes one square from each of the three quarters without the black square; every quarter is then a 4-by-4 board with one taken square, which the method of Problem 2 covers with five pieces: $1 + 4 \cdot 5 = 21$. The 16-by-16 board: cut it into four 8-by-8 quarters, place the central piece, and cover each quarter with the method of this problem: $1 + 4 \cdot 21 = 85$ pieces. The same step takes 16 to 32, 32 to 64, and so on. A child who says “you keep doing the same thing one size up” has understood an induction.



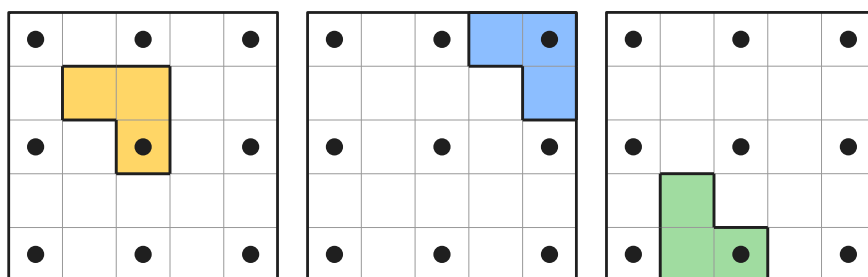
The covering of the printed 8-by-8 board: the central piece, then the central piece of each quarter, then the 2-by-2 blocks.

Problem 4. Checks on the nine squares whose row and column are both odd (the corners, the middles of the sides, the center); X on the sixteen others. The figure gives coverings for a corner, a middle of a side and the center; the other six check squares are these turned or flipped.



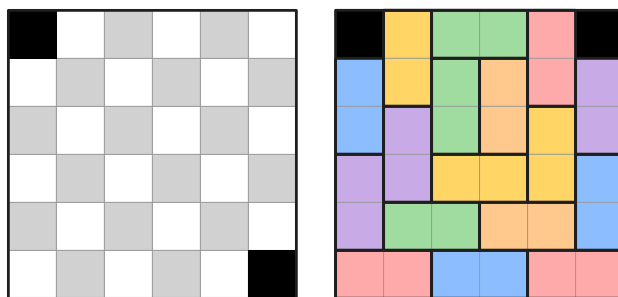
The 5-by-5 board: the three kinds of good position, and the map of all nine.

Problem 5. Every count is 0 or 1, never 2. An L-piece fits inside some 2-by-2 block of the board, and every 2-by-2 block contains exactly one dotted square, because among two consecutive rows exactly one is odd, and the same for columns. A covering has 8 pieces (24 squares), so it covers at most 8 dots; there are 9; so some dot is not covered, and the only uncovered square is the black one. Hence the black square is on a dot. Problem 4 showed that every dot works, so the answer is complete: exactly the nine dotted squares.



Wherever an L-piece lies, its 2-by-2 block holds exactly one dot.

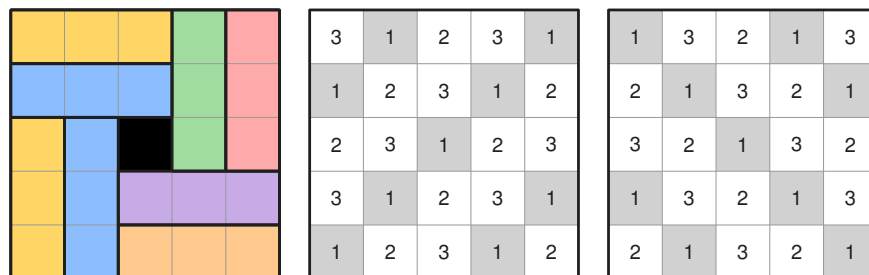
Problem 6. Opposite corners: impossible. The marking is the checkerboard: every domino covers one shaded and one unshaded square, because squares that share a side differ. The full board has 18 and 18; the two corners have the same shade; 16 shaded and 18 unshaded remain; 17 dominoes would need 17 of each. The checkerboard is the only marking that does the job: any two squares sharing a side must differ, which forces the pattern once one square is decided. Corners of the same side: 17 and 17 remain, and the covering exists (two lying dominoes finish the top row; the 5-by-6 rectangle below is filled row by row).



The 6-by-6 board: the checkerboard count, and a covering for corners of the same side.

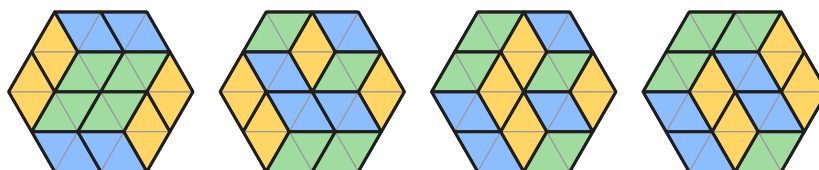
Problem 7. Only the center works. Numbering: write 1, 2, 3, 1, 2, 3, ... along each diagonal, so that each row reads 1 2 3 1 2 shifted by one step from the row above; then any three squares in a row or a column carry 1, 2 and 3 in some order. One number appears on 9 squares and the other two on 8 squares each (the shaded squares in the figure). A covering uses 8 pieces, so 8 squares of each number; the black square must carry the number that appears 9 times. That leaves 9 candidates. The numbering along the

other diagonals leaves 9 other candidates, and the only square in both sets is the center. The pinwheel covering shows that the center works.



The pinwheel covering; the two diagonal numberings with their 9-square classes shaded.

Problem 8. There are 20 different coverings of the big hexagon by 12 rhombi, and every one of them has exactly 4 rhombi of each kind. What the children notice: the colored pictures look like cubes stacked in the corner of a 2-by-2-by-2 box, and the counts are always 4, 4, 4 (four tops, four left faces and four right faces are visible). Why there are always 4 of kind 1, the tall rhombi whose two triangles share a horizontal edge: a tall rhombus takes a down-pointing triangle from one horizontal row and the up-pointing triangle right above it from the next row, while rhombi of kinds 2 and 3 lie inside a row and take one up and one down triangle of that row. The bottom row has 2 up and 3 down triangles, so exactly one of its down triangles must be covered by a tall rhombus reaching into the second row. The second row has 3 up and 4 down; one up triangle is already taken from below; the rest pair up inside the row except for 2 down triangles, which must be covered by tall rhombi reaching into the third row. The third row has 4 up and 3 down; two up triangles are taken from below, so exactly one down triangle reaches into the top row. The top row has 3 up and 2 down; one up triangle is taken from below and the remaining 2 and 2 pair up inside the row. So every covering has $1 + 2 + 1 = 4$ tall rhombi, and turning the hexagon by one sixth of a turn gives 4 for each other kind.



Four of the twenty coverings, colored by kind.

Hints, in order. Problem 1: “Which piece can cover the square next to the black one? Try every way it can sit.” Problem 2, for the method: “The black square is in one quarter. Look at the other three quarters.” then “Is there one piece that touches all three of those quarters at once?” Problem 3: “Which quarter has the black square? Give each other quarter one taken square with a single piece.” then “Now each quarter is Problem 2.” Problem 4: “Corner, middle of a side, center first. Then the square next to a corner.” A child who tries the square next to a corner will fail; after three tries, mark X and move on; Problem 5 explains it. Problem 5: “Draw a 2-by-2 box around your piece. How many dots are inside any 2-by-2 box?” then “Eight pieces: how many dots can they cover at most?” Problem 6: “Mark so that two squares that share a side are never both marked.” Problem 7: “Number the top row 1 2 3 1 2. The square under the 1 cannot be 1: what can it be?” then “Where does the number that appears 9 times live?” Problem 8: “Start at a corner of the hexagon: the corner triangle has only one neighbor.” For a fourth covering: “Make the top corner a different kind of rhombus.”

If the table stalls on the method in Problem 2, let the children finish their own coverings of the six small boards first (they will succeed every time), then show the first picture of the Problem 2 figure and ask

what that one piece does to each quarter. If the counting in Problem 5 does not land, run it as a dialogue: “How many dots? How many pieces? At most how many dots per piece? So at most how many dots covered? Which dot is left?”

If the table finishes early. Problems 7 and 8. Then: which of these boards, with no black square, can be covered by L-pieces? 2-by-3, yes. 3-by-3, no: each piece covers at most one corner, and three pieces cannot cover four corners. 3-by-4, yes (two 2-by-3 blocks). 3-by-5, no: whichever piece covers the top-left square, the first two columns can only be completed by exactly two pieces, so the pieces use up the columns two at a time and the number of columns must be even. 6-by-6, yes (six 2-by-3 blocks). Then: a 7-by-7 board with one black square (16 pieces) can be covered for every position of the black square; let them find one.

The share (seven minutes)

Order: K–1, then grades 2–3, then grades 4–5; two minutes each; a child holds up the page and an adult helps with the words.

K–1 shows shape F with green triangles on it: “7 gray, 5 white; every rhombus eats one of each; two grays are left over.” Next to it shape D, equal counts, covered.

Grades 2–3 show the 6-by-6 board with opposite corners: “16 shaded, 18 unshaded, 17 dominoes.” If there is time, the loop on the 4-by-4 board: “when the two black squares have different shades it always works, because the loop breaks into two even pieces.”

Grades 4–5 show the 8-by-8 board covered by L-pieces and say how they would do 16-by-16; then the 5-by-5 board with dots: “9 dots, 8 pieces, each piece covers at most one dot.”

Closing line for the organizer: “Three tables, three kinds of pieces, one idea: find two kinds of cells that every piece uses equally, count them, and the count decides. And one more idea from the oldest table: a method that works at one size and gives you the next size for free.”

The mathematics behind the session, for the two mathematicians

Perfect matchings. A domino tiling of a region of the square grid is a perfect matching of its grid graph, and the checkerboard coloring is the bipartition. The count argument (equal numbers of black and white cells) is the crudest instance of Hall’s condition; the full condition, every set S of black cells has at least $|S|$ white neighbors, is necessary and sufficient, and a connected region with equal counts can still fail it: a 1-by-4 strip with one extra cell attached below its second cell and one attached above its third cell has three cells of each color and no tiling, because the two attached cells force their only neighbors and isolate the two ends. Deciding whether a region has a domino tiling is bipartite matching, hence polynomial; for simply connected regions Thurston (1990) gave an algorithm linear in the area, built on height functions. The loop argument of the grades 2–3 table is Gomory’s proof that removing any black and any white square from the 8-by-8 board leaves a tileable board; any Hamiltonian cycle of the grid graph serves.

Counting tilings. Domino tilings of the 2-by- n strip are counted by the Fibonacci numbers 1, 2, 3, 5, 8, 13, . . . , by the first-column recursion the children find. The 4-by-4 board has 36 domino tilings, the 6-by-6 board 6728 and the 8-by-8 board 12 988 816; Kasteleyn and, independently, Temperley and Fisher (1961)

found a product formula for the m -by- n board, the starting point of the dimer model in statistical physics. Lozenge tilings of the hexagon with sides a, b, c (the grades 4–5 Problem 8 has $a = b = c = 2$) correspond to plane partitions in an $a \times b \times c$ box, which is what the stack-of-cubes picture shows; MacMahon's formula counts them, giving 2, 20 and 980 for the regular hexagons of side 1, 2 and 3. The constant counts of each lozenge kind (bc , ca and ab ; here 4, 4, 4) follow from the row-by-row argument given above, or from the picture: each column of the box shows exactly one top face. Random tilings of large hexagons and of Aztec diamonds show the arctic circle phenomenon (Jockusch, Propp and Shor; Cohn, Larsen and Propp, 1998): outside an inscribed ellipse the tiling is frozen.

Trominoes. The induction for 2^n -by- 2^n boards is Golomb's (1954). Chu and Johnsonbaugh (Mathematics Magazine, 1986) classified the deficient square boards: for n not divisible by 3, every n -by- n board with one cell removed can be tiled by L-trominoes, except for $n = 5$, where exactly the nine cells with odd row and odd column work; for n divisible by 3 the number of cells is not divisible by 3. The 7-by-7 board therefore works for every removed cell (also checked by computer for this packet, as was every count in this guide). Rectangles: a 3-by- n board can be tiled by L-trominoes only for even n , by the two-columns-at-a-time argument above, and the 3-by-3 board fails because each piece covers at most one corner. For straight trominoes the two diagonal 3-colorings are the standard tool: on the 8-by-8 board with one cell removed they leave exactly four admissible cells, $(3, 3)$, $(3, 6)$, $(6, 3)$ and $(6, 6)$, and each of these can in fact be tiled (Golomb's board-with-a-monomino problem); on the 5-by-5 board they leave only the center.

Beyond colorings. Conway and Lagarias (1990) introduced tiling groups and showed that some impossibilities cannot be proved by any coloring argument: the triangular region T_N of side N in the triangular lattice can be tiled by tribones (three cells in a line) only when $N \equiv 0, 2, 9$ or $11 \pmod{12}$, and this obstruction is invisible to colorings. Thurston (1990) recast these invariants as height functions, which also give the fast domino algorithm mentioned above.

Hardness and undecidability. Deciding whether a given finite region of the grid can be tiled by L-trominoes is NP-complete (Moore and Robson, 2001; the same holds for 2-by-2 squares), so no quick general test like the one for dominoes exists unless $P = NP$. For the infinite plane, Berger (1966) proved that no algorithm decides whether a finite set of square tiles with colored edges (Wang tiles) tiles the plane; the proof produced the first aperiodic tile sets, ancestors of the Penrose tiles and of the aperiodic monotile found in 2023 by Smith, Myers, Kaplan and Goodman-Strauss.

Where the tables' ideas go. The $K-1$ up-and-down count is the bipartition of the honeycomb lattice, the dual of the triangular grid, on which lozenge tilings are dimer configurations. The grades 2–3 Fibonacci recursion is the simplest transfer-matrix computation. The grades 4–5 induction is the model case of a recursive construction, and the 5-by-5 board the model case of a small exception explained by a counting invariant.